AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 18 · STUDENT LESSON

Dataset and benchmark limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Count the sources. Name the setting. Keep unknowns visible.

Audit duplicate weighting and benchmark coverage without inventing independent units or unmeasured performance.

Keep one habit: Source contract → ten-row identity ledger → native setting-coverage chart → slice/pooled probes → metadata audit → independent twelve-row case.

MAIN LESSON · SECTION 1

Audit what the benchmark represents before extending its score

> Rows, source units and intended settings answer different questions.

Learning goals

I can explain the **mechanism** and terminology of dataset and **benchmark** limitations.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve experimental units and evaluation scope

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting response

- Recall Sensitivity analysis. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain dataset and **benchmark** limitations using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of dataset and **benchmark** limitations

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Separate identity, coverage and correctness

> Each missing property needs evidence rather than a plausible assumption.

Begin with the benchmark’s boundary

A benchmark score belongs to a particular set of tasks, sources, labels and collection conditions. Ten rows need not represent ten distinct sources, and a clean indoor photo score need not describe shaded outdoor photos. Before interpreting a result, count the declared source units and identify the intended setting that is absent.

Keep missing performance unknown rather than converting it to zero or assuming the measured score transfers.

Original benchmark explanation

A **benchmark** measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Complete source and coverage contract

Audit item	Complete supplied contract
Main source arrangement	Ten rows: three copies of S1 plus seven other distinct source IDs S2–S8
Distinct source count	One repeated source + seven others = eight
Duplicate weighting	S1 contributes three rows; each other source contributes one
Extra rows beyond one per source	Two; do not confuse three copies with three extra copies
Independence	Different source IDs do not prove independent sampling
Separate picture	Indoor 9/10; outdoor initially unmeasured
Changed picture	Four reserved outdoor cases supplied, two correct
Unassessed properties	Source independence, label validity, eligibility, future representativeness and raw per-source outcomes

Given and missing facts

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace ten rows back to eight source identities

> Three copies represent one source and two extra rows beyond a one-per-source count.

Original eligibility and identity procedure

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

Complete illustrative row-to-source ledger

Declared source ID	Row IDs in illustrative arrangement	Row weight
S1	R1, R2, R3	3 of 10
S2	R4	1 of 10
S3	R5	1 of 10
S4	R6	1 of 10
S5	R7	1 of 10
S6	R8	1 of 10
S7	R9	1 of 10
S8	R10	1 of 10
Total	Ten rows; eight distinct IDs	S1 receives triple row weight

Declared source ID	Row IDs in illustrative arrangement	Row weight
Evidence limit	IDs illustrate supplied arrangement	No labels or independent sampling inferred

Preserve exclusions, assumptions and visual explanation

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Dataset and benchmark limitations — evidence trace

INPUT AND SCOPE

Ten benchmark rows include three copies of one source and seven other unique sources.

CHECK OR CALCULATION

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

CONCLUSION AND LIMIT

Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict what four supplied outdoor cases add

> New outdoor evidence narrows one gap without proving future performance.

Main task and response

- Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict changed outdoor slice

- Change: Add four reserved outdoor cases, of which two are correct.

▼ **Check after predicting**

Prediction: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read measured and missing settings beside their denominators

> Keep the main source-count example separate from the photo coverage example.

Main distinct-source count and scope

Ten **benchmark** rows include three copies of one source and seven other unique sources.

Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate indoor/outdoor facts

A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

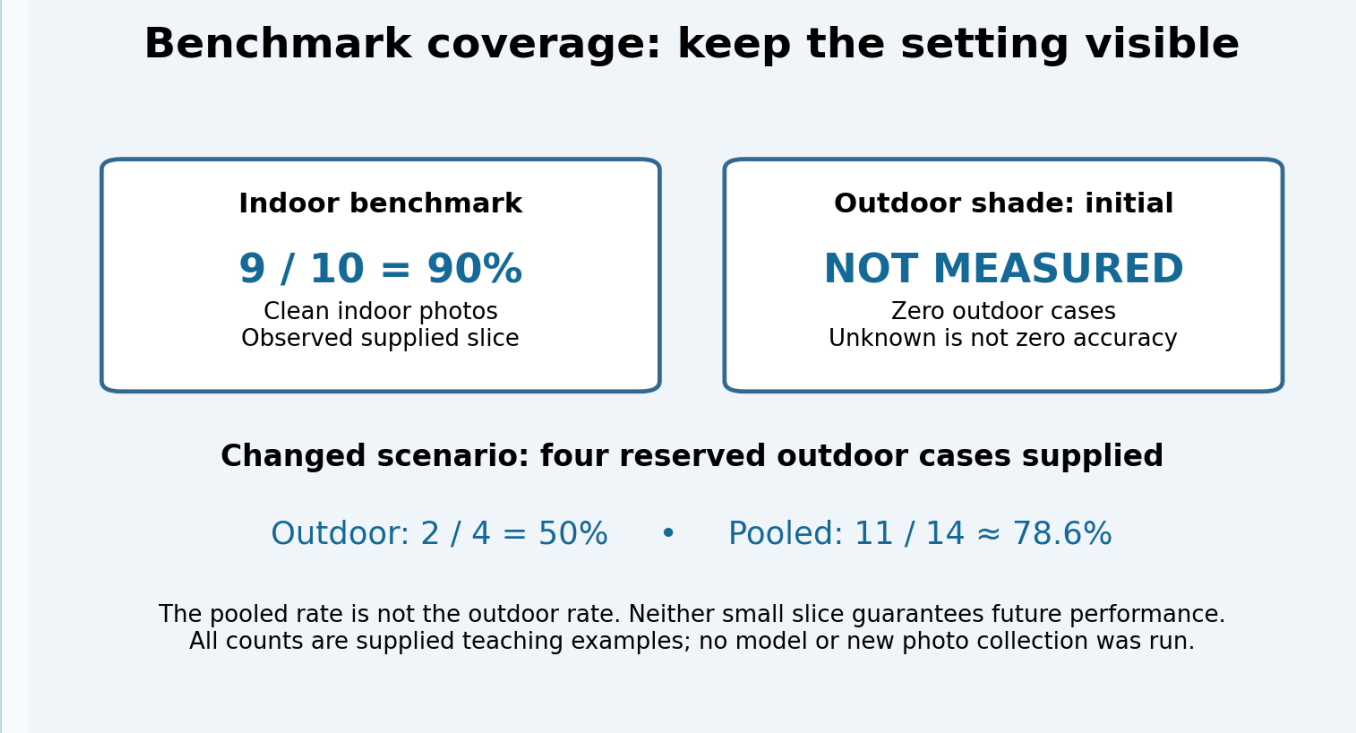
Indoor **benchmark**: 9/10 correct. No outdoor cases are present. The same model is intended for both.

The picture's original ten photos are all clean indoor images: nine correct gives 90% on that supplied slice. There are no outdoor cases, so outdoor accuracy is unmeasured. In the changed teaching scenario, four reserved outdoor cases are supplied and two are correct,

giving 50% for that slice. Pooling both settings gives 11/14, approximately 78.6%, but that pooled rate is not outdoor performance.

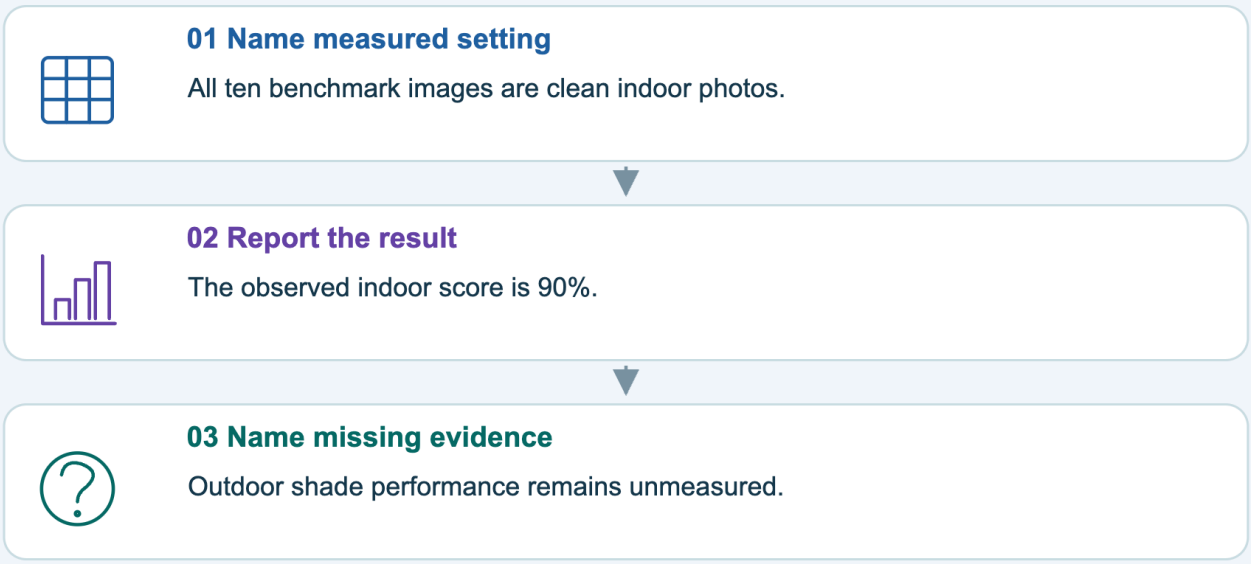
The lower observed outdoor slice does not identify the cause, establish significance or prove all future outdoor images will behave similarly. No new photos or model outputs were actually collected for this lesson.

See measured, missing and added slices



AI ACADEMY / AI-7/18 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name measured setting	All ten benchmark images are clean indoor photos.
2. Report the result	The observed indoor score is 90%.
3. Name missing evidence	Outdoor shade performance remains unmeasured.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Complete slice and pooled-rate comparison

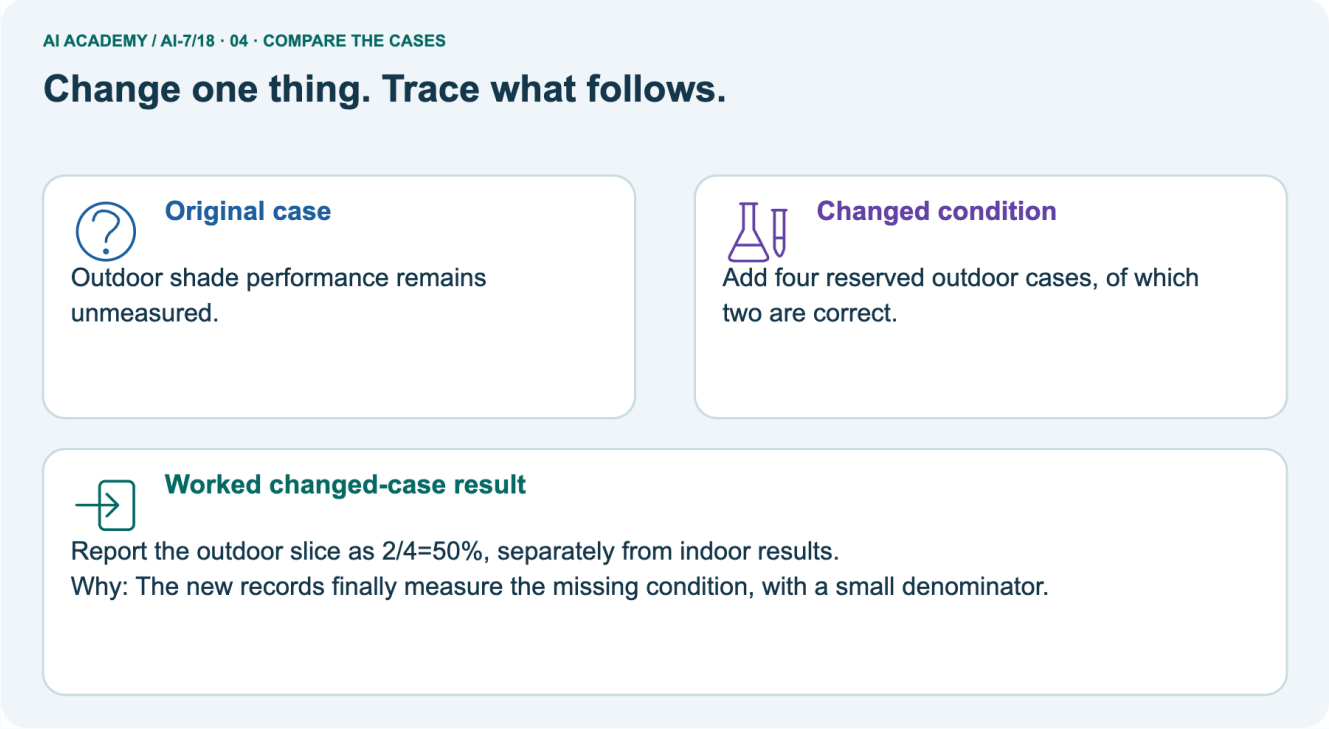
Setting	Original supplied evidence	Changed supplied evidence
Indoor	9/10 = 90%	Unchanged 9/10 = 90%
Outdoor shade	No cases; rate unknown	2/4 = 50%
Pooled descriptive rate	9/10, entirely indoor	11/14 ≈ 78.6%
Appropriate outdoor report	Unmeasured	2/4 in the supplied outdoor slice
Unsupported extension	Assume outdoor also 90%	Use pooled 78.6% as outdoor accuracy

Setting	Original supplied evidence	Changed supplied evidence
Remaining limits	Coverage absent	Small slice, sampling and labels still need inspection

Predict adding four outdoor cases

- Change one thing: Add four reserved outdoor cases, of which two are correct. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining coverage limits

- Neither slice estimates all future photos; inspect sampling, labels and grouping.

- Read the labelled picture: Report the indoor **benchmark** with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dataset and benchmark limitations

The facts we will use

AI ACADEMY / AI-7/18

Benchmark conditions

	Correct	Total
Indoor benchmark	9	10
Added outdoor slice	2	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/18

Dataset and benchmark limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name measured setting

All ten benchmark images are clean indoor photos.

↓

2. Report the result

The observed indoor score is 90%.

↓

3. Name missing evidence

Outdoor shade performance remains unmeasured.

CHANGE ONE THING

Add four reserved outdoor cases, of which two are correct.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair duplicate weighting and unsupported coverage claims

> Preserve original records and explain every exclusion or unassessed condition.

Original limitations and repair setup

Benchmark success does not establish performance on a deployment population missing from its design.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete identity and coverage probes

Benchmark audit	Checked interpretation
Three copies of one source plus seven others	Eight source IDs, not ten independent sources
Three copies	Two extra rows beyond one copy, not three extra rows
Source IDs differ	Identity distinction does not prove sampling independence
No outcome labels supplied per source	Do not calculate an invented deduplicated accuracy
Exclude duplicates after inspecting outcomes	Disclose analysis change; preserve original results and declared unit rule
No outdoor cases	Unknown performance, not zero accuracy
Outdoor 2/4 and pooled 11/14	Different denominators and target descriptions
High indoor result	No automatic outdoor-fitness conclusion

Benchmark audit	Checked interpretation
Observed slice difference	Does not by itself establish cause or significance

Write a supported repair

- A report omits this qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise identifying units and missing settings

> Reserve the twelve-row source arrangement for a first independent attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example “Dataset and **benchmark** limitations”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual check

- Use the worked case for Dataset and **benchmark** limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Report the indoor **benchmark** with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve selected source-count case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete metadata and coverage audit

> Distinguish counted identities from verified sampling properties.

Current investigation

Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

Complete executable audit

```
"""Source-identity and coverage audit on explicitly fictional metadata/summaries."""
from collections import Counter
from fractions import Fraction

def audit_sources(rows):
 if not isinstance(rows,(list,tuple)) or not rows:raise ValueError('nonempty records
required')
 if any(not isinstance(r,dict) or not isinstance(r.get('row_id'),str) or not r['row_id']
or not isinstance(r.get('source_id'),str) or not r['source_id'] for r in rows):raise
ValueError('row and source IDs required')
 if len({r['row_id'] for r in rows})!=len(rows):raise ValueError('row IDs must be unique')
 counts=Counter(r['source_id'] for r in rows)
 return {'rows':len(rows),'unique_source_ids':len(counts),'source_weights':dict(counts),
         'extra_rows_beyond_one_per_source':len(rows)-len(counts)}
# Metadata construction illustrates the stated three-copy/eight-source arrangement.
rows=[{'row_id':f'R{i+1}','source_id':s} for i,s in
enumerate(['S1','S1','S1','S2','S3','S4','S5','S6','S7','S8'])]
before=[r.copy() for r in rows];main=audit_sources(rows)
assert main['rows']==10 and main['unique_source_ids']==8 and main['source_weights']
['S1']==3
assert main['extra_rows_beyond_one_per_source']==2 and rows==before
indoor=Fraction(9,10);outdoor_initial=None;outdoor_added=Fraction(2,4)
pooled_changed=Fraction(9+2,10+4)
assert indoor==Fraction(9,10) and outdoor_added==Fraction(1,2) and
pooled_changed==Fraction(11,14)
print('Supplied-source metadata arrangement:',main)
print('Indoor supplied rate:',indoor,'initial outdoor rate:',outdoor_initial)
print('Changed supplied outdoor slice:',outdoor_added,'pooled descriptive
count:',pooled_changed)
```

```
print('Do not substitute the pooled 11/14 for outdoor 2/4, or call unmeasured outdoor performance zero.')
print('IDs distinguish declared source identities; they cannot prove independent sampling, eligibility or correct labels. No rows are silently deleted, no outcomes per source invented, and no real photo collection/model evaluation occurs.')
```

Expected counts and limits

1. Predict the eight distinct source IDs, the repeated source's three-row weight and the two extra rows. The code audits illustrative metadata matching the supplied arrangement without deleting records. It computes separate indoor, added-outdoor and pooled rates, keeping initial outdoor performance unknown.

   IDs cannot verify independent sampling or correct labels, and no per-source outcome is supplied for a deduplicated accuracy. This local arithmetic does not collect photos or evaluate a model. Keep the independent twelve-row arrangement for a first attempt.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: count source units in twelve rows

> State one coverage concern without inventing the missing evidence.

Independent source arrangement

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

> **My independent procedure, result and limitation**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save row identities, source grouping and unassessed settings

> Retain original records and explain exclusions before interpreting a score.

Project and artifact

Add a labelled dataset and **benchmark** limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of dataset and **benchmark** limitations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local calculation or paper trace

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain what a benchmark measured and what it missed

> Identity, labels, sampling and coverage remain separate checks.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of dataset and **benchmark** limitations

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Benchmark	A declared task dataset and scoring procedure
Duplicate unit	Repeated representation of the same underlying entity
Coverage gap	A relevant condition absent from evaluated data

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Dataset and benchmark limitations — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ten benchmark rows include three copies of one source and seven other unique sources.	Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.	Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

By the end, I can	How I will show it
Explain the mechanism and terminology of dataset and benchmark limitations	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.
- Which labelled step matches this detail? Outdoor shade performance remains unmeasured.
- 1. Report the result
- 2. Name measured setting
- 3. Name missing **evidence**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Name missing **evidence** Outdoor shade performance remains unmeasured. A **benchmark** represents particular tasks and collection conditions.

High performance on it cannot automatically establish fitness for a different deployment setting.

Changed-case question and optional hint

- Changed-case practice: Add four reserved outdoor cases, of which two are correct. Which conclusion is supported by the supplied facts?
- 1. Report the outdoor slice as 2/4=50%, separately from indoor results.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Report the outdoor slice as 2/4=50%, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

Repair and retain the record

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

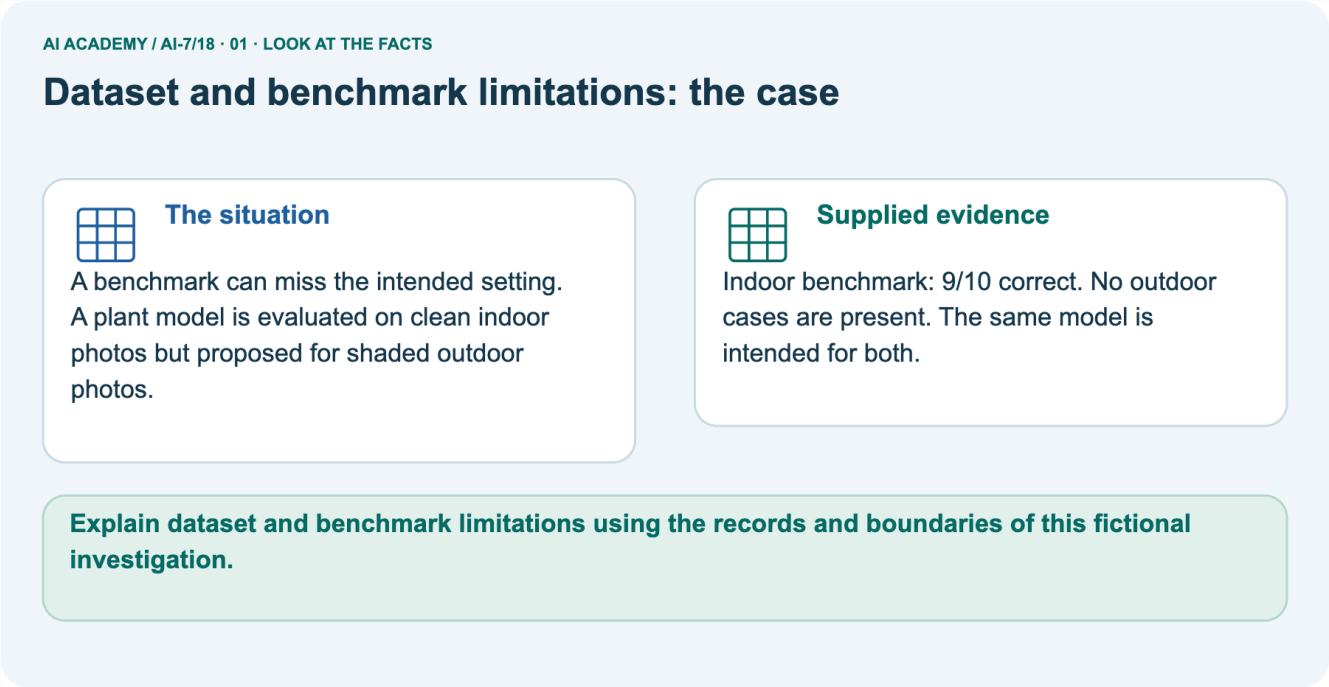
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Dataset and benchmark limitations: the case



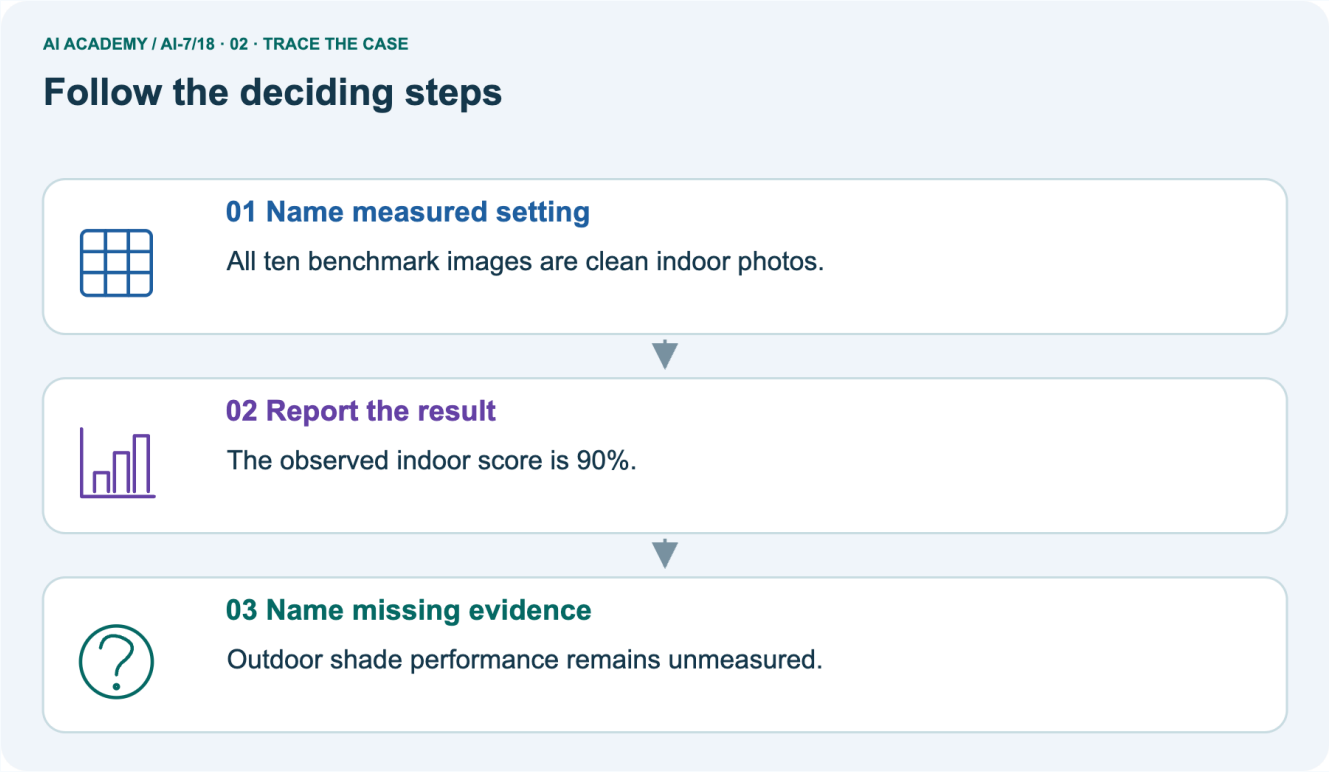
Follow the labelled picture with the matching explanation. Open the vector diagram

A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos. Supplied facts: Indoor **benchmark**: 9/10 correct. No outdoor cases are present.

The same model is intended for both.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

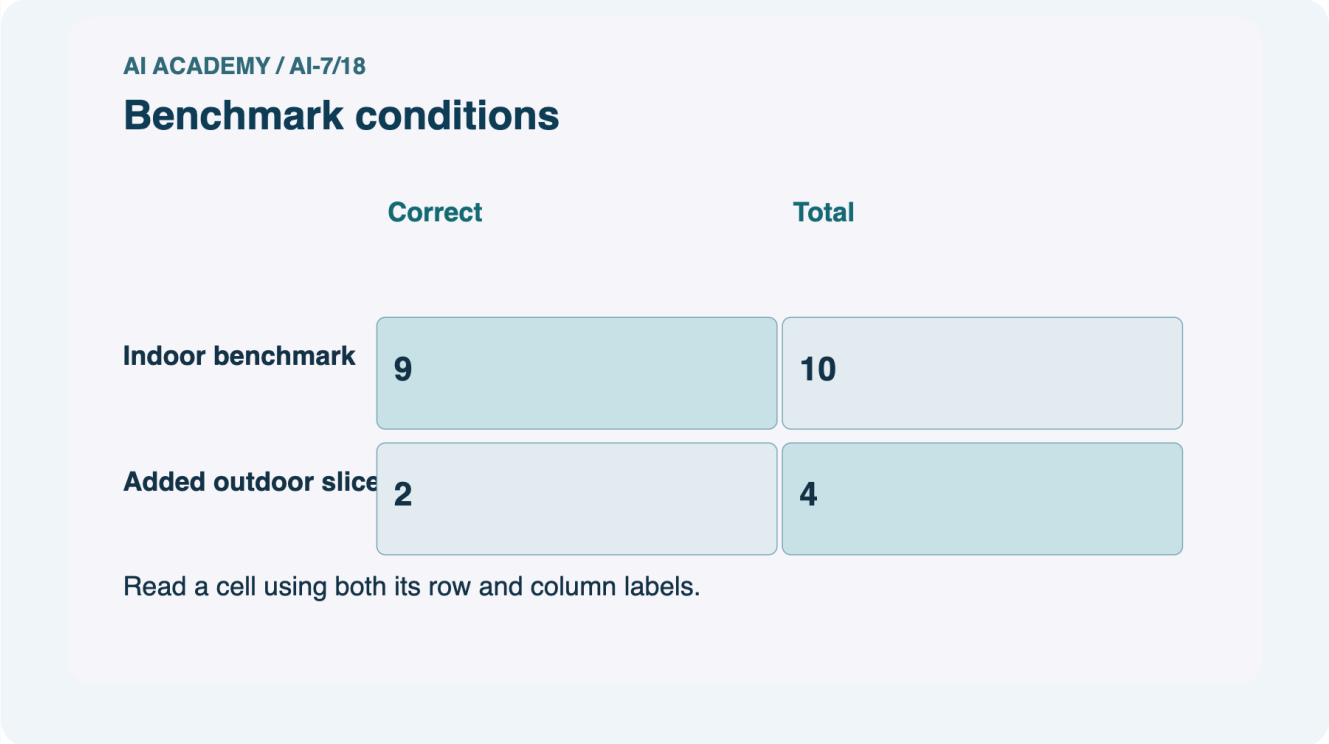


Follow the labelled picture with the matching explanation. Open the vector diagram

Name measured setting: All ten **benchmark** images are clean indoor photos. → Report the result: The observed indoor score is 90%. → Name missing **evidence**: Outdoor shade performance remains unmeasured.

Point to each numbered stage and explain which recorded **input** it uses.

03. Benchmark conditions

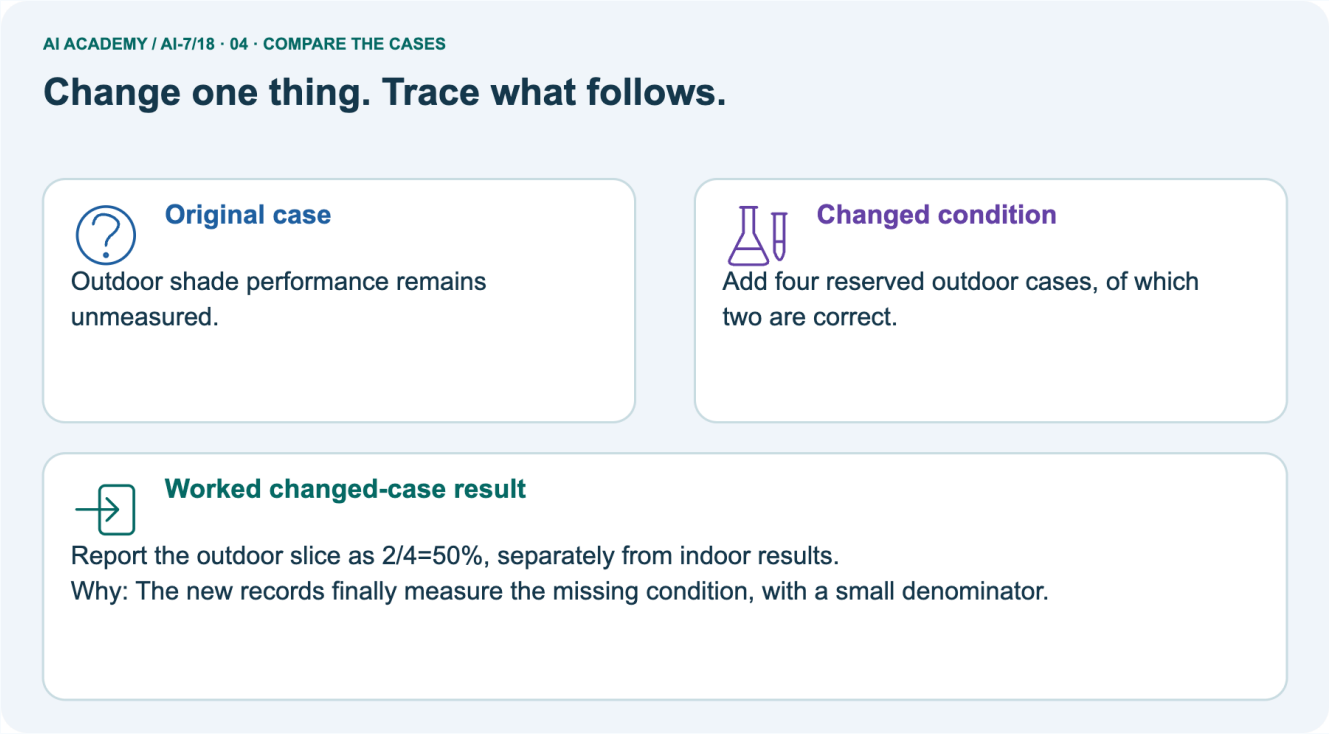


Follow the labelled picture with the matching explanation. Open the vector diagram

Benchmark conditions. Indoor **benchmark**: Correct = 9, Total = 10. Added outdoor slice: Correct = 2, Total = 4.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



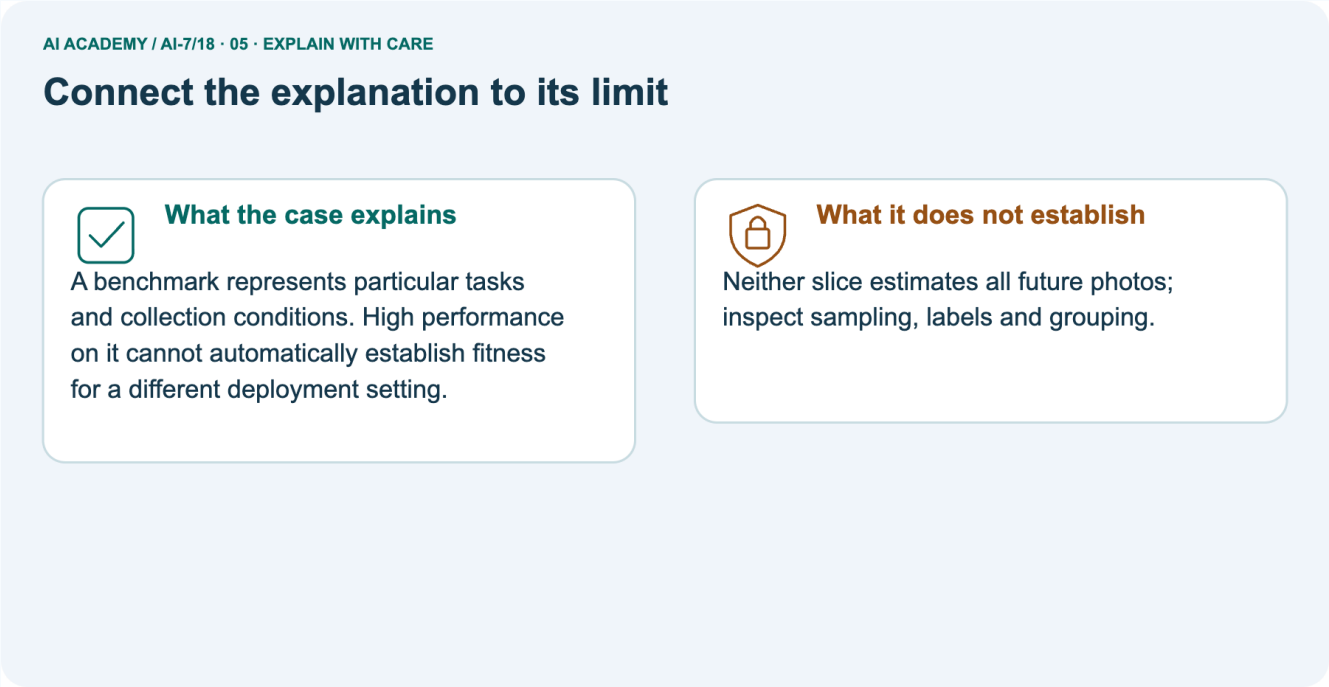
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Outdoor shade performance remains unmeasured. Changed condition: Add four reserved outdoor cases, of which two are correct. Expected result: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



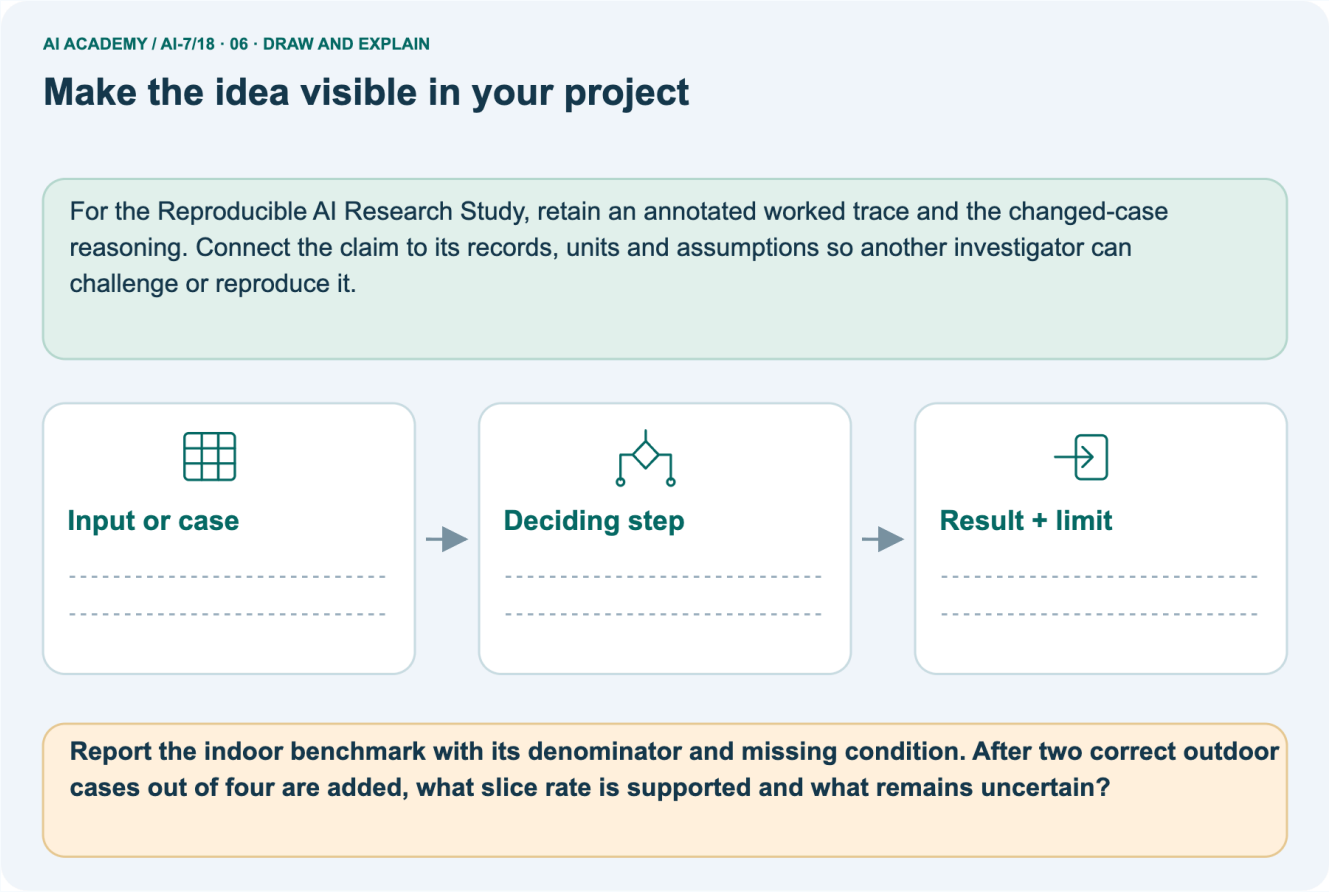
Follow the labelled picture with the matching explanation. Open the vector diagram

A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Limit: Neither slice estimates all future photos; inspect sampling, labels and grouping.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Report the indoor **benchmark** with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Dataset and benchmark limitations: the case

02. Follow the deciding steps

03. Benchmark conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.